

10.- Si las rectas L y R se cortan, determinar las coordenadas de su punto de intersección.

$$L: \begin{cases} x - 4 = 1 - y = \frac{z - 3}{2} \end{cases} \quad R: \vec{r} = (5 - n, 6 - 2n, -3 + 2n) \quad n \in \mathbb{R}$$

$$L: \begin{cases} x = 4 + m \\ y = 1 - m \\ z = 3 + 2m \end{cases} \quad m \in \mathbb{R}$$

$$R: \begin{cases} x = 5 - n \\ y = 6 - 2n \\ z = -3 + 2n \end{cases} \quad n \in \mathbb{R}$$

Iguando abscisas y

$$4 + m = 5 - n$$

$$1 - m = 6 - 2n$$

$$m + n = 1 \dots (1)$$

$$+ \quad m + 2n = 5 \dots (2)$$

$$3n = 6$$

$$\boxed{n = 2}$$

Sustituyendo n en (1)

$$m + (2) = 1$$

$$\boxed{m = -1}$$

Iguando cotas y
sustituyendo m, n

$$3 + 2m = -3 + 2n$$

$$3 + 2(-1) = -3 + 2(2) \rightarrow 3 - 2 = -3 + 4 \rightarrow$$

$$\boxed{1 = 1}$$

Sustituyendo m en las ecs. par. L

$$x = 4 + (-1) = 3$$

$$y = 1 - (-1) = 2 \rightarrow \underline{I(3, 2, 1)}$$

$$z = 3 + 2(-1) = 1$$

$\therefore \exists I$

11- Calcular la distancia entre las rectas M y N de ecuaciones

$$M: \vec{m} = (4 + 3t)\vec{i} + (-3 - 2t)\vec{j} + (-2t)\vec{k} \quad t \in \mathbb{R}$$

$$N: \begin{cases} \frac{x-2}{3} = z + s \\ y = 2 \end{cases} \quad \begin{aligned} P &= (4, -3, 0) \rightarrow \vec{p} = (4, -3, 0) \\ \vec{u} &= (3, -2, -2) \\ Q &= (2, 2, -s) \rightarrow \vec{q} = (2, 2, -s) \\ \vec{v} &= (3, 0, 1) \end{aligned}$$

$$d = \frac{|(\vec{q} - \vec{p}) \cdot \vec{u} \times \vec{v}|}{|\vec{u} \times \vec{v}|}$$

$$d = \frac{|[(2, 2, -s) - (4, -3, 0)] \cdot (3, -2, -2) \times (3, 0, 1)|}{|(3, -2, -2) \times (3, 0, 1)|}$$

$$d = \frac{|(-2, 5, -s) \cdot (-2, -9, 6)|}{|(-2, -9, 6)|}$$

$$d = \frac{|(-2)(-2) + 5(-9) + (-s)(6)|}{\sqrt{(-2)^2 + (-9)^2 + 6^2}}$$

$$d = \frac{|4 - 45 - 30|}{\sqrt{4 + 8 + 36}}$$

$$d = \frac{|-71|}{\sqrt{44}} = \frac{71}{\sqrt{44}} = 6.450$$

\$188.80

\$308 + \$231

369.5

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -2 & -2 \\ 3 & 0 & 1 \end{vmatrix} = (-2, -9, 6)$$

12.- Determinar la distancia entre las rectas L y R expresadas por

$$L: \begin{cases} x = -1 \\ y = -3z + 9 \end{cases}$$

$$P(-1, 0, 3) \rightarrow \vec{p} = (-1, 0, 3)$$

$$R: \vec{r} = (2, -4 + 3t, -9 - t) \quad t \in \mathbb{R}$$

$$\rightarrow Q(2, -4, -9)$$

$$\vec{u} = (0, 1, -1/3)$$

$$\vec{v} = (0, 3, -1)$$

$$\vec{q} = (2, -4, -9)$$

$$d = \frac{|(\vec{q} - \vec{p}) \times \vec{u}|}{|\vec{u}|}$$

$$d = \frac{|[(2, -4, -9) - (-1, 0, 3)] \times (0, 3, -1)|}{|(0, 3, -1)|}$$

$$d = \frac{|(3, -4, -12) \times (0, 3, -1)|}{\sqrt{3^2 + (-1)^2}}$$

$$d = \frac{|(40, 3, 9)|}{\sqrt{10}}$$

$$d = \frac{\sqrt{40^2 + 3^2 + 9^2}}{\sqrt{10}}$$

$$d = \frac{\sqrt{1600 + 9 + 81}}{\sqrt{10}} = \frac{\sqrt{1690}}{\sqrt{10}} = \sqrt{\frac{1690}{10}} = \sqrt{169} = 13 \text{ u.}$$

$$(\vec{q} - \vec{p}) \times \vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -4 & -12 \\ 0 & 3 & -1 \end{vmatrix} = \begin{vmatrix} 3 & -4 & -12 \\ 0 & 3 & -1 \\ 3 & -4 & -12 \end{vmatrix} = (40, 3, 9)$$